

MUTHURAJA M

muthuraja05980@gmail.com | +91 9344058085 | linkedin.com/in/muthuraja93 | github.com/Muthuraja997 | leetcode.com/u/muthuraja05980

SUMMARY

AI Engineer with hands-on experience building multi-agent AI systems, RAG pipelines, and production-ready ML applications. Proficient in Python, SQL, LLM integration, and prompt engineering, with a track record of delivering scalable, data-driven solutions.

SKILLS

Technical: Python, Machine Learning, Deep Learning, NLP, LLM, RAG, SQL, OOP, FastAPI, Git/GitHub, AWS (Bedrock, Lambda, Textract, S3, Agent Core)

Soft Skills: Problem Solving, Adaptability, Team Collaboration

EXPERIENCE

AI/ML Engineering Intern — Aivar

Oct 2025 – Aug 2026

- Designed and implemented an enterprise-grade RAG chatbot using Amazon Bedrock (embeddings + generation) with a Qdrant vector database for semantic search and retrieval.
- Built a document ingestion pipeline using AWS Textract and Docling for structured and unstructured data processing.
- Developed FastAPI backends for AI-powered conversational workflows and integrated LiveKit (WebRTC) for real-time voice communication, with pipeline orchestration via Pipecat.
- Built a Post Call Analysis (PCA) pipeline for AI voice agents, extracting structured data from call transcripts and enriching it with contextual knowledge using RAG.

PROJECTS

Multi-Agent Medical Assistant Platform

Python, FastAPI, LLM, RAG, LangGraph

- Built a multi-agent AI system for X-ray, skin lesion, and brain tumor detection with real-time appointment scheduling via Jitsi Meet.
- Developed a RAG pipeline for context-aware medical Q&A and a voice-enabled psychology agent using Speech-to-Text.

Academia Nexus — AI-Powered Student Intelligence Platform

Python, MySQL, Gemini API, MCP

- Architected a multi-agent system on Model Context Protocol (MCP) for career prediction, interview prep, scholarship discovery, and learning analytics.
- Integrated Google Gemini API for context-aware question generation, career recommendations, and AI-powered test generation with skill-gap analysis.

Voice Coding Assistant

TypeScript, Python, FastAPI, WebSockets, Llama-3.3-70B

- Built a VS Code extension with a WebSocket-based FastAPI backend enabling natural-language voice/text coding queries and workspace file operations.
- Integrated Llama-3.3-70B-Instruct with tool calling for project-aware reasoning and automated file operations (create, read, update, delete, search).

Sentiment Analysis Using YouTube Watch History

Python, Pandas, NLTK

- Built an NLP pipeline to extract video metadata via the YouTube Data API and classify user content preferences through sentiment polarity analysis.

EDUCATION

B.Tech, AI & Machine Learning — Sri Shakthi Institute of Engineering and Technology, Coimbatore 2022 – 2026

CGPA: 7.93/10

Higher Secondary Certificate — Government Higher Secondary School, Virudhunagar

2021 – 2022

Percentage: 80%

CERTIFICATIONS

AWS Machine Learning Associate (AWS) | Python (GUVI) | LLM Workshop (IIT Madras) | Pandas (Udemy)

ACHIEVEMENTS

- Finalist, ABB EngineeredX Hackathon
- Secured 1st Place, ICPC 2025 (College Level)